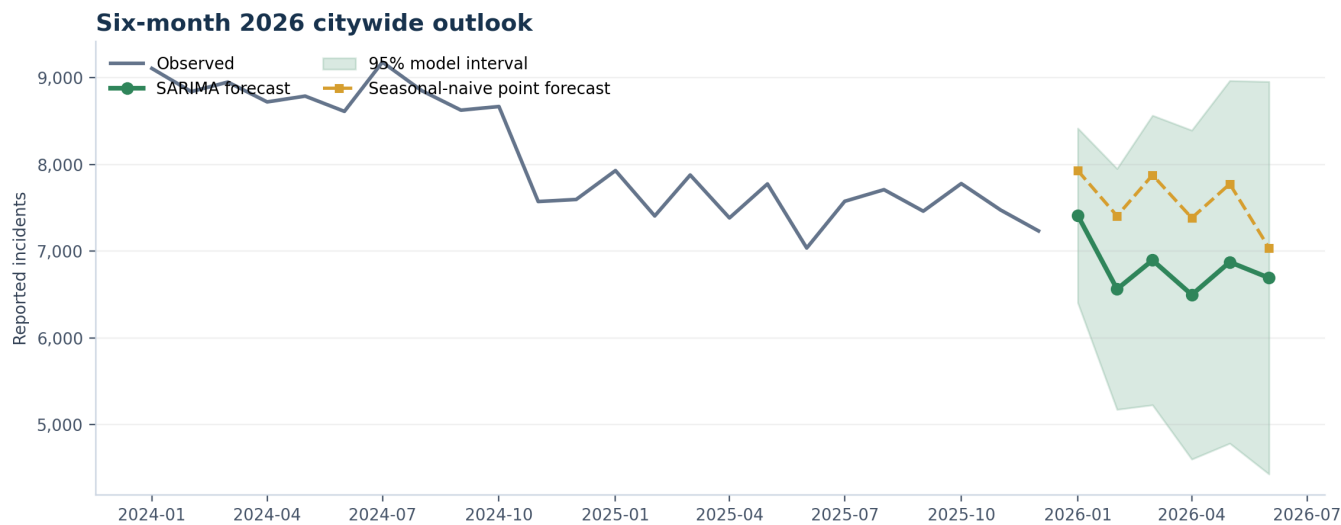


Reported-Incident Forecasting

A reproducible, uncertainty-aware public-data analysis and six-month citywide outlook

Professional Portfolio Report

Study window: January 2018 through December 2025
Forecast horizon: January through June 2026
Data source: DataSF, Police Department Incident Reports (wg3w-h783)



Historical observations, SARIMA forecast, 95% model interval, and seasonal-naive point baseline.

Sileshi Hirpa | Data Science & Business Analytics

Python · SARIMA · ETS · Streamlit · Plotly · Parquet · Rolling-origin validation

1. Executive summary

942,427	96	6	0.80
recorded incidents	complete months	backtest origins	best mean MASE

This project converts eight complete years of San Francisco Police Department reported-incident records into validated analytical aggregates, a comparative time-series evaluation, a six-month citywide forecast, and an interactive Streamlit product.

SARIMA produced the lowest mean MASE (0.80), narrowly ahead of ETS (0.81), with mean MAPE of 5.78% and MAE of 432 reported incidents. The result is a modest empirical advantage, not evidence of universal model superiority.

Primary conclusion

Reported incident volume changed materially after 2020 and does not simply return to its pre-2020 level. The validated 2026 outlook centers near 6,822 incidents per month, but must be interpreted as a model-based citywide estimate with uncertainty - not a causal claim, neighborhood risk score, or operational recommendation.

Decision value

- Establishes a transparent, repeatable benchmark for citywide reported-incident volume.
- Keeps model performance, uncertainty, data provenance, and limitations visible together.
- Separates historical neighborhood/category exploration from the validated citywide monthly forecast.

2. Business understanding and analytical scope

The analytical objective is to estimate aggregate monthly reported-incident volume and communicate the result responsibly. The product is designed for portfolio demonstration and directional planning context, not real-time public-safety operations.

Scope boundaries

- Unit of analysis: recorded SFPD incident reports; counts are not population-adjusted rates.
- Historical study window: 2018-01 through 2025-12, containing 96 continuous months.
- Forecast target: total citywide monthly reported incidents for a six-month horizon.
- Exploratory dimensions: year, DataSF Analysis Neighborhood, incident category, weekday, and hour.
- Excluded claims: causation, unreported crime, individual risk, neighborhood safety ranking, and patrol allocation.

Data provenance and controls

The pipeline retrieves DataSF dataset wg3w-h783 through deterministic pagination, normalizes the schema, validates monthly continuity, materializes versioned Parquet aggregates, and records row counts and SHA-256 hashes in metadata. The committed deployment artifacts eliminate a near-million-row runtime download while preserving rebuild instructions.

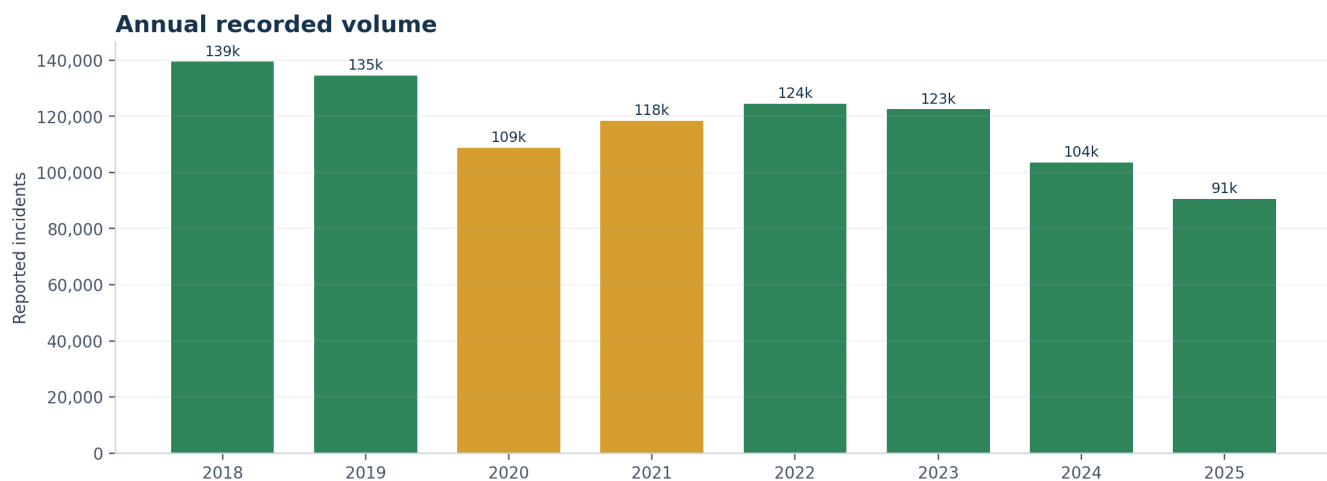


Figure 1. Annual recorded volume. The highlighted 2020-2021 period is treated as a structural disruption rather than a causal estimate.

3. CRISP-DM methodology

CRISP-DM phase	Implementation in this project
1. Business understanding	Define a defensible aggregate forecasting question, intended users, decision boundaries, and non-goals.
2. Data understanding	Audit coverage, schema, missingness, continuity, category concentration, spatial patterns, and weekday-hour rhythms.
3. Data preparation	Normalize dates and dimensions, validate the complete monthly index, and generate versioned analytical aggregates.
4. Modeling	Fit seasonal naive, additive Holt-Winters ETS, and SARIMA(1,1,1)(1,1,1,12).
5. Evaluation	Use six expanding-window origins with a fixed three-month horizon; compare MASE, MAE, RMSE, MAPE, and WAPE.
6. Deployment	Serve historical analysis and forecast outputs in Streamlit, supported by CI checks, tests, model-card documentation, and reproduction commands.

Methodological strengths

- Time-ordered evaluation prevents future observations from leaking into earlier training windows.
- A seasonal-naive comparator tests whether complexity adds value beyond a credible baseline.
- MASE is the primary selection metric because it is scale-free and benchmark-relative; supporting metrics preserve interpretability.
- The final forecast is refit only after model selection, using the complete 2018-2025 monthly series.

4. Data understanding and exploratory findings

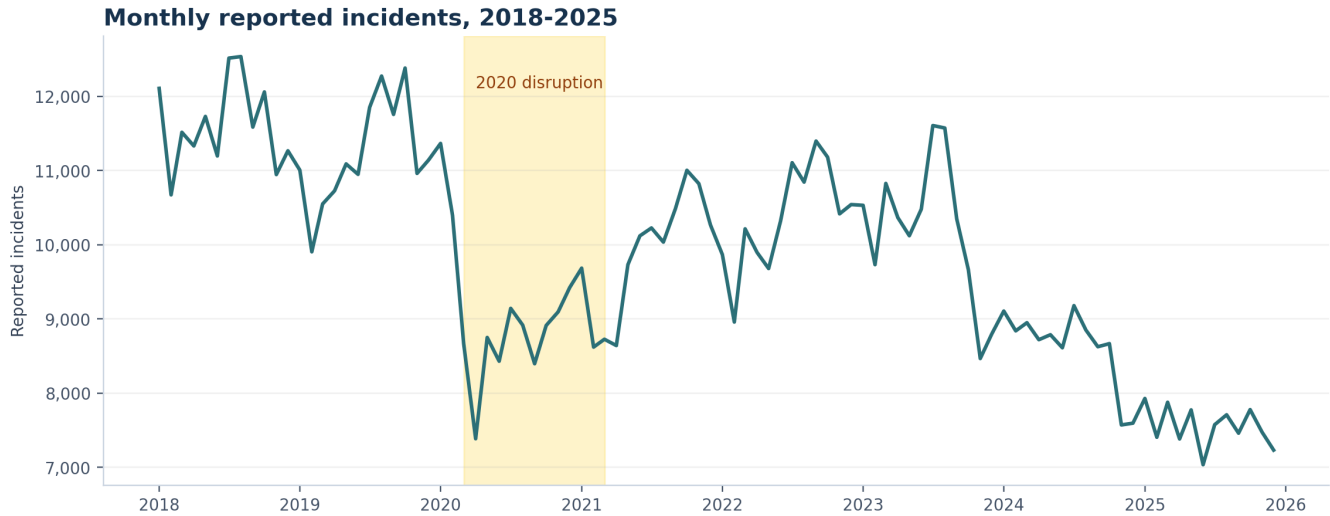


Figure 2. Monthly reported-incident counts. The shaded interval marks the onset and immediate aftermath of the 2020 disruption.

Key findings

- The series exhibits recurring seasonal structure, supporting explicit annual seasonality in all candidate models.
- The 2020 period marks a material structural disruption, and the subsequent series does not simply return to its pre-2020 level.
- Recorded incidents concentrate in Mission, Tenderloin, SoMa, Financial District, and Bayview, but raw totals reflect exposure, land use, enforcement, and reporting behavior.
- Weekday-by-hour patterns reveal temporal rhythms that monthly aggregates cannot show.

Interpretation discipline

Exploratory concentration is descriptive. A high raw count is not equivalent to a high resident risk rate because the analysis does not normalize for population, foot traffic, daytime population, visitor activity, or enforcement intensity.

5. Forecast evaluation

Candidate models were evaluated across six expanding-window forecast origins, each with a fixed three-month horizon. Metrics below are means across historical folds.



Model	MASE	MAPE	MAE	RMSE	WAPE
Seasonal naive	1.97	14.14%	1,062	1,113	14.12%
ETS (Holt-Winters)	0.81	5.81%	434	490	5.78%
SARIMA	0.80	5.78%	432	482	5.76%

Selection judgment

SARIMA recorded the lowest mean MASE, but the difference from ETS is only 0.01. Accordingly, the evidence supports SARIMA as the selected model for this release while recognizing ETS as a statistically competitive challenger. The seasonal-naive baseline is materially weaker across all reported metrics.

These values summarize historical backtests and are not guaranteed future accuracy. Continued evaluation should monitor error by horizon and interval calibration as new complete months arrive.

6. Rolling-origin consistency

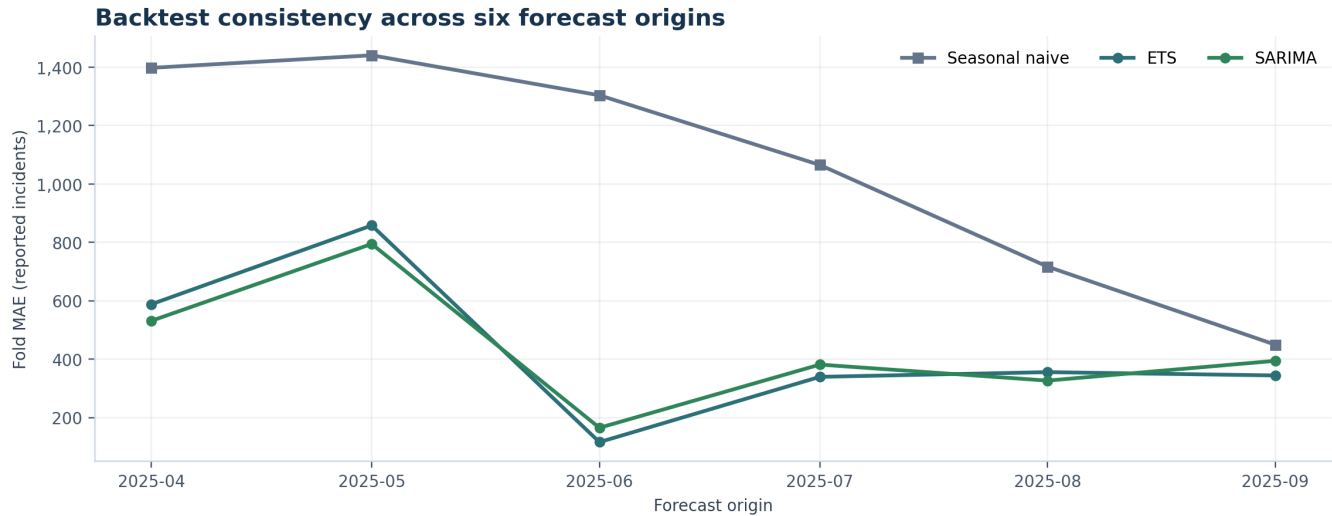


Figure 4. Fold-level MAE across the six stored 2025 forecast origins. The evaluation window does not span the pandemic, so pandemic shading is intentionally omitted.

Why this is stronger than a single score

The fold-level view shows whether the selected model's advantage is persistent or concentrated in one favorable holdout. SARIMA and ETS remain closely competitive, while the seasonal-naive baseline is consistently less accurate over the stored origins.

The available backtest origins run from April through September 2025. Claiming that this chart proves robustness across the 2020 pandemic period would exceed the evidence. Structural-break sensitivity is therefore documented as a limitation and monitoring concern rather than presented as a validated result.

7. Uncertainty-first 2026 outlook

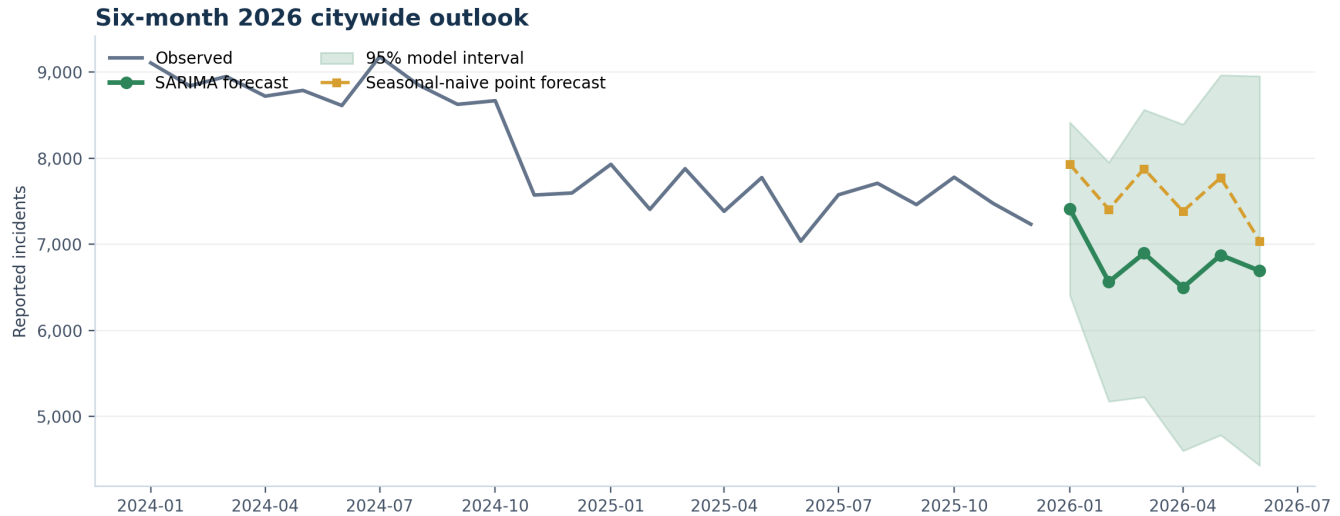


Figure 5. Six-month SARIMA forecast with model-based 95% prediction intervals and a seasonal-naive point baseline.

The point forecast centers near 6,822 reported incidents per month over the six-month horizon. Prediction intervals widen uncertainty communication beyond a single deterministic line, but they capture model uncertainty only; they do not account for every future policy, event, reporting, or economic shock.

The seasonal-naive series is shown as a point baseline only because a directly comparable baseline interval is not stored. The report does not imply that SARIMA has a tighter interval without estimating both intervals under a common method.

Appropriate use

- Use the forecast for directional citywide volume context and analytical benchmarking.
- Review new complete months against the forecast, baseline, and interval before retraining.
- Investigate source revisions, category changes, or pipeline anomalies before attributing forecast error to behavior.

Inappropriate use

- Do not infer neighborhood-level or person-level risk from the citywide forecast.
- Do not treat incident counts as estimates of all crime, including unreported events.
- Do not use the model as a causal evaluation or patrol-allocation tool.

8. Reporting behavior and domain insight

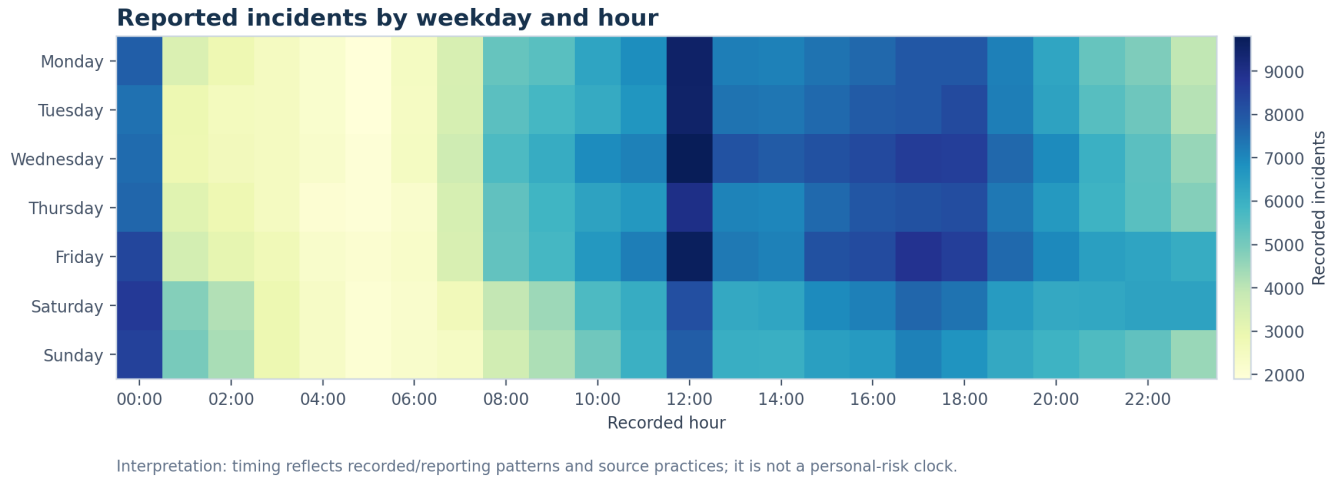


Figure 6. Aggregate recorded-incident volume by weekday and hour across the complete 2018-2025 window.

The heatmap makes recurring temporal patterns visible without presenting them as a personal-safety clock. Recorded timestamps can reflect event timing, delayed reporting, enforcement activity, and source-system practices.

Interpretation guide

- Read color intensity as recorded volume, not probability of victimization.
- Compare recurring reporting rhythms; do not infer causation from time-of-day concentration.
- Use category-specific drilldowns in the application when a domain question requires more context.

9. Data provenance and product engineering

Reproducible data-to-product lineage

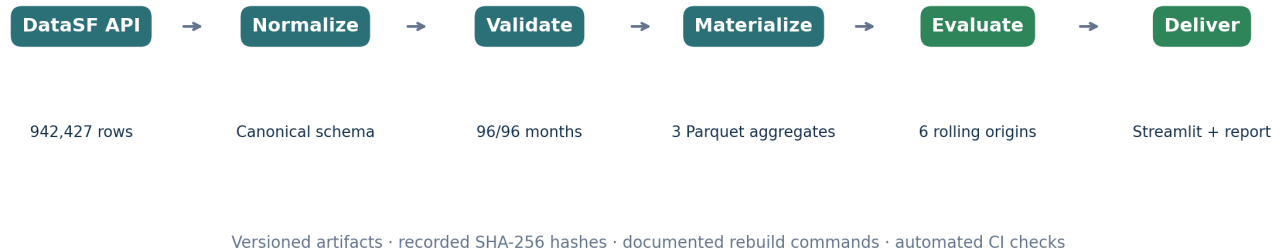


Figure 7. Reproducible data-to-product lineage.

The Streamlit application separates filterable historical analysis from the validated citywide monthly forecast. Users can explore historical trends by year range, neighborhood, and incident category, while model comparison and uncertainty remain visible in the forecast view.

Engineering controls

- Deterministic API pagination and canonical schema normalization.
- Versioned Parquet aggregates with metadata, row counts, and cryptographic hashes.
- Automated CI checks, schema validation, forecasting tests, and compile checks.
- Generated model card, documented reproduction commands, and deployment-ready lightweight artifacts.
- Clear separation between tested production modules and the narrative portfolio notebook.

10. Limitations, monitoring, and responsible use

Risk area	Implication
Measurement	Records reflect reporting, enforcement, classification, and source-system practices; they do not represent all incidents.
Structural change	The 2020 disruption may reduce the stability of relationships learned from earlier observations.
Aggregation	Citywide monthly totals conceal local heterogeneity and do not describe individual or neighborhood risk.
Predictors	The models exclude weather, events, policy changes, economic factors, and reporting delays.
Intervals	Prediction intervals are model-based and require ongoing empirical calibration.

Monitoring protocol

- Re-run the rolling-origin backtest when new complete months arrive.
- Compare SARIMA against ETS and seasonal naive; do not assume the incumbent remains best.
- Track MAE, MASE, and interval coverage by forecast horizon.
- Review material data revisions and schema changes before retraining.
- Document model, data, metric, and artifact changes in the model card and release history.

11. Reproducibility and conclusion

Reproduce locally

Clone the repository, install Python 3.11 dependencies with `uv sync --extra dev`, and launch the committed aggregates with `uv run streamlit run app.py`. Rebuild from DataSF with `uv run sf-incidents-build`, regenerate evaluation outputs with `uv run sf-incidents-evaluate`, and run quality checks with `uv run ruff check .` and `uv run pytest`.

Canonical project resources

- Repository: github.com/sileshith/sf-incident-forecasting
- Live application: sf-incident-forecasting.streamlit.app
- Portfolio: hirpadata.com
- Data source: data.sfgov.org/d/wg3w-h783

Conclusion

The project demonstrates a complete analytical lifecycle: defensible problem framing, validated data preparation, exploratory analysis, benchmarked time-series modeling, uncertainty-aware communication, deployment, and monitoring. Its strongest contribution is not a single forecast value; it is the combination of reproducibility, comparative evidence, product usability, and explicit limits on interpretation.

Prepared by Sileshi Hirpa

Data Science · Business Analytics · Mathematical Foundations